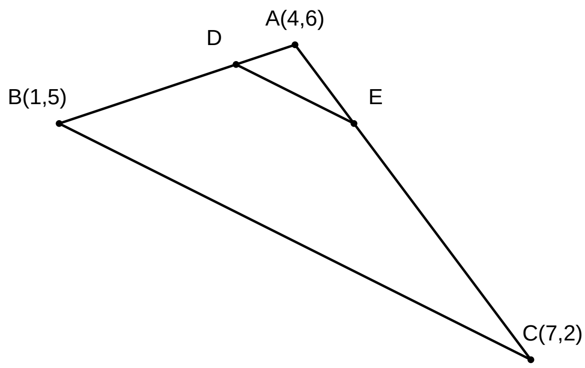
Coordinate LA Smoke — Solutions

---

1. 1. The vertices of $riangle ABC$ are $A(4, 6)$, $B(1, 5)$, and $C(7, 2)$. A line is drawn to intersect sides AB and AC at points D and E respectively, such that $\frac{AD}{AB} = \frac{AE}{AC} = \frac{1}{4}$. Find the coordinates of points D and E and show that the area of $riangle ADE$ is $\frac{1}{16}$ of the area of $riangle ABC$.



Answer:

$$Coordinates of D are (3.25, 5.75), E are (4.75, 5), and the ratio of areas is \frac{1}{16}.$$

Solution:

1. Use the section formula to find coordinates of D on AB dividing it in ratio $1 : 3$.
2. Use the section formula to find coordinates of E on AC dividing it in ratio $1 : 3$.
3. Calculate the area of $riangle ABC$ using the determinant formula for coordinates.
4. Calculate the area of $riangle ADE$ using the same formula.
5. Compare the two areas to verify the ratio is $1 : 16$.

Derivation:

$$D = \left(\frac{1(1) + 3(4)}{1 + 3}, \frac{1(5) + 3(6)}{1 + 3} \right) = (3.25, 5.75)$$

$$E = \left(\frac{1(7) + 3(4)}{1 + 3}, \frac{1(2) + 3(6)}{1 + 3} \right) = (4.75, 5)$$

$$Area(ABC) = \frac{1}{2}|4(5 - 2) + 1(2 - 6) + 7(6 - 5)| = \frac{1}{2}|12 - 4 + 7| = 7.5$$

$$Area(ADE) = \frac{1}{2}|4(5.75 - 5) + 3.25(5 - 6) + 4.75(6 - 5.75)| = \frac{1}{2}|3 - 3.25 + 1.1875| = 0.46875$$

$$\frac{Area(ADE)}{Area(ABC)} = \frac{0.46875}{7.5} = \frac{1}{16}$$

Coordinate Geometry — Section Formula and Area of Triangle